Hiran Sarkar

67, Badridas Temple Street
Kolkata, 700004
(+91) 7980543067
hiransarkar2001@gmail.com

Website
aroundstar
Google Scholar

Area(s) of interest: Rigid and deformable 3D/4D reconstruction & generation, implicit deep learning, human pose estimation, generative modeling, video understanding, few-shot/incremental/open-set learning & object detection

EDUCATION

Simon Fraser University - *Doctor of Philosophy (PhD)* SEPT 2026 - AUG 2031

School of Computing Science, advised by Prof. Hao (Richard) Zhang

Netaji Subhash Engineering College - *Bachelor's of Technology* JULY 2019 - JULY 2023

Major in Computer Science and Engineering

CGPA: 9.14 / 10

EXPERIENCE

Technical University of Munich, CAMP - *Research Intern* APRIL 2023 - MAR 2026

Worked on **generalized continuous dynamic scene representation** using NeRF and Neural ODE advised by *Prof. Benjamin Busam*. The paper is accepted in **CVPR 2026.**

- We learn the underlying physics of scenes with deterministic dynamics, using Neural Radiance Field that learns the spatial information and neural ODE that learns the temporal dynamics.

  This setup enables:

  - *Fine-grained interpolation and long-term extrapolation of the scenes:* Predicting the evolution of the dynamics beyond the training data.
  - *Generalization to different initial conditions:* For eg., varying the initial position and velocity of dynamic objects. Trained on sparse values of initial conditions, the model can generate novel trajectories for unseen configurations, marking its ability to learn the underlying dynamics of a scene beyond seen trajectories.

Sony Research India - *Research Intern* APRIL 2023 - MAR 2026

- Worked on **open-set object detection** where we addressed the high unknown misclassification problem. We introduced a clip based clustering based module that forms clusters in the semantic space forming a clear boundary separation especially between classes of close semantic proximity. The paper is accepted in **WACV 2024** as an oral presentation.
- Worked on a survey paper on **few-shot object detection (FSOD)** that covers different variants of the field including standard FSOD, generalized FSOD, incremental FSOD, open-set FSOD, and domain adaptive FSOD.
- Worked on **generative modeling**: text conditioned image to video generation, LCM-LoRA for faster diffusion model inference and real to anime video generation.

Indian Institute of Science, Bangalore, Vision & AI Lab - *Research Intern* JUNE 2021 - MAY 2022

I have worked on **domain adaptation** and **domain generalization** using self-supervision across multiple settings under *Prof. Venkatesh Babu*.

- Worked on unsupervised source-free domain adaptation technique using a subsidiary pretext task which acts as an aid to help attain domain invariance thereby minimizing domain discrepancies. The paper is accepted in **ECCV 2022**.
- Developed an unsupervised non-source-free domain adaptation technique using a Bag-of-VisualWords (BoW) like representation. We looked into the tradeoff between negative transfer risk and domain invariance exhibited at different layers of the network. The paper is accepted in **NeurIPS 2022**.

Teamcognito, Kolkata - *Machine Learning Intern* OCT 2020 - JAN 2021

- Worked on a speech recognition model using Recurrent Neural Network Transducer. Trained on the Common-Voice dataset.
- Worked on *Malware Detection on IoT devices based on traffic Meta-Data*

RESEARCH PAPERS

Node-RF: Learning Generalized Continuous Space-Time Scene Dynamics with Neural ODE-based NeRFs [Link]
Computer Vision and Pattern Recognition (CVPR) 2026
Hiran Sarkar, Liming Kuang, Yordanka Velikova, Benjamin Busam

Beyond Few-shot Object Detection: A Detailed Survey [Link]
arXiv 2025
Hiran Sarkar*, Vishal Chudasama*, Pankaj Wasnik, Vineeth Balasubramanian, Jayateja Kalla

Open-Set Object Detection By Aligning Known Class Representations [Link]
Winter Conference on Applications of Computer Vision (WACV) 2024. ***(Oral Presentation)***
Hiran Sarkar, Vishal Chudasama, Naoyuki Onoe, Pankaj Wasnik, Vineeth Balasubramanian

Subsidiary Prototype Alignment for Universal Domain Adaptation [Link]
Neural Information Processing Systems (NeurIPS) 2022
Jogendra N. Kundu*, Suvaansh Bhambri*, Akshay R. Kulkarni*, **Hiran Sarkar**, Varun Jampani, Venkatesh Babu R.

Concurrent Subsidiary Supervision for Unsupervised Source-Free Domain Adaptation [Link]
European Conference on Computer Vision (ECCV) 2022
Jogendra N. Kundu*, Suvaansh Bhambri*, Akshay R. Kulkarni*, **Hiran Sarkar**, Varun Jampani, Venkatesh Babu R.

PROJECTS

Image Style Transfer - [Link]

Implemented the paper *Rethinking and Improving the Robustness of Image Style Transfer* in TensorFlow. This paper examines why pre-trained ResNet networks hinder Image Style Transfer compared to VGG, despite ResNet's superiority in tasks like classification. It employs the L-BFGS optimizer to accelerate training.

Cartoon GAN - [Link]

This project aims to transform real-world photos into cartoon-style images using a generative adversarial network (GAN) comprising a generator and a discriminator. It employs two types of losses: adversarial loss (discriminator loss), which guides the generator to achieve the desired manifold transformation, and content loss (perceptual loss), which ensures the preservation of image content during stylization.

Emojify - [Link]

Returns the emoji that matches the input text, using LSTM network.

Neural Machine Translation - [Link]

Performed neural machine translation from English to Spanish using Bahdanau Attention mechanism and deployed it using Flask.

OPEN SOURCE

Winter of Code at DeepFusionAI, Remote - *Open Source Developer* DEC 2020 - FEB 2021

- Worked on Social Distance Detector using MobileDets.

Contributor at TensorFlow

- Fixed various bugs and added a feature in the main TensorFlow GitHub repository.
    - Added the axis argument in tensorflow.keras.losses.categorical_crossentropy() and in tensorflow.keras.losses.binary_crossentropy(). **[Link]**
    - Fix for Conv1DTranspose, Conv2DTranspose, Conv3DTranspose layers when filter=0 by adding a ValueError. **[Link]**

Awards

Sony Research India Annual Awards - FY 23: Best paper-Gold Category for *Open-Set Object Detection By Aligning Known Class Representations*. **[Link]**

5G Hackathon: Phase 1 winner, among Top 100 ideas on our project *Fertilizer Management System* in 5G Hackathon 2021. Our team won a prize of Rs 1,00,000. **[Link]**

Prayas 2020: A national level inter-college project competition. Our team stood 3rd for building real-time object detection on mobile devices.

Hult Prize: Our team stood second in Hult Prize OnCampus at Netaji Subhash Engineering College 2020.